Revision — parallelism and perpendicularity in space

The Grade 10 statements, revisited with a year of solids behind them.

LESSON 63–64 | 2 × 40 MIN | GEOMETRY 10–11 REVIEW | SPATIAL REVIEW

LESSON CLOCK

15' | 25' | 25' | 20' | 15'

The statements	15 min
Angles, both ways	25 min
Distances	25 min
Choosing a method	20 min
Homework	15 min

LEARNING OBJECTIVES

- State the positions of lines and planes, and the criteria for parallelism and perpendicularity.
- Use the three-perpendiculars theorem to reduce a space problem to a plane one.
- Compute angles between lines, between a line and a plane, and between planes.
- Choose between the synthetic and the vector method for each question.

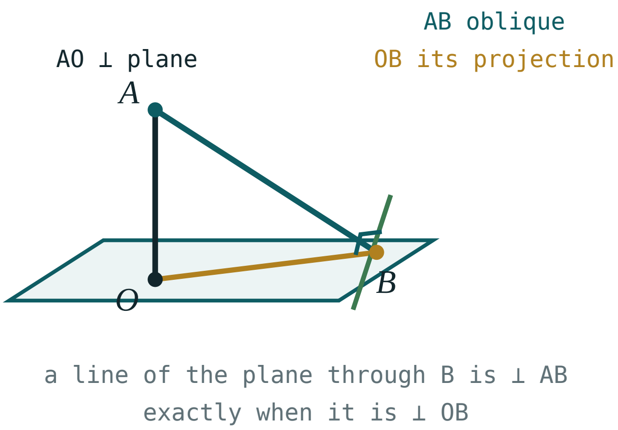
TEXTBOOK

National	pp. 325–334
Cambridge	Pure Mathematics 2 & 3, pp. 281–284
Workbook	Revision set G3

01 Explanation

The statements

STATEMENT	CONDITION
a line is parallel to a plane	it is parallel to some line **in** the plane
two planes are parallel	two **intersecting** lines of one are parallel to two of the other
a line is perpendicular to a plane	it is perpendicular to two **intersecting** lines of it
two planes are perpendicular	one contains a line perpendicular to the other
three perpendiculars	a line in a plane ⊥ an oblique ⇔ ⊥ its projection

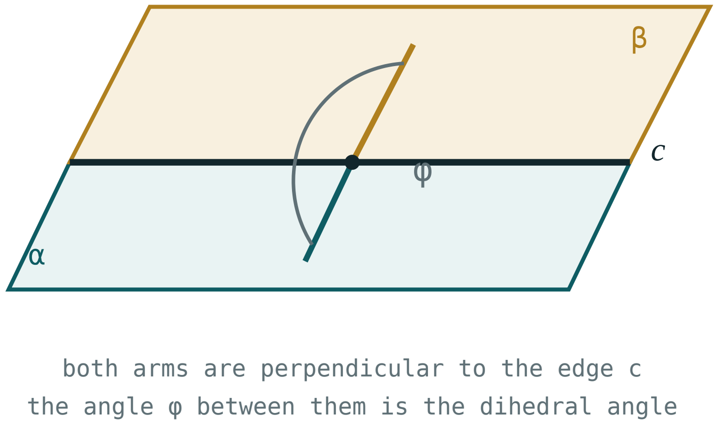


The theorem that turns a space problem into a plane one.

> “INTERSECTING” APPEARS TWICE, AND IS ESSENTIAL BOTH TIMES
>
> Two *parallel* lines of one plane parallel to two of another prove nothing — a line and a plane through it satisfy that with no parallelism at all. The same word carries the same weight in the perpendicularity criterion.

Angles, both ways

ANGLE	SYNTHETIC METHOD	VECTOR METHOD
two lines	translate one to meet the other	$\cos\theta = \dfrac{\lvert d_1 \cdot d_2 \rvert}{\lvert d_1 \rvert \lvert d_2 \rvert}$
line and plane	angle with the projection	$\sin\theta = \dfrac{\lvert d \cdot n \rvert}{\lvert d \rvert \lvert n \rvert}$
two planes	the linear angle at the edge	$\cos\theta = \dfrac{\lvert n_1 \cdot n_2 \rvert}{\lvert n_1 \rvert \lvert n_2 \rvert}$



The linear angle — both rays perpendicular to the edge.

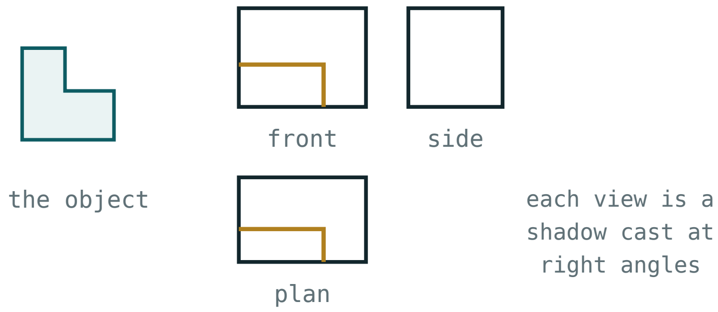
The same question, both ways. In a cube $ABCDA_1B_1C_1D_1$ of edge 1, the angle between AC_1 and the base:

SYNTHETIC	VECTOR
projection $AC = \sqrt{2}$, height 1, $\tan\theta = \dfrac{1}{\sqrt{2}}$	$d = (1,1,1)$, $n = (0,0,1)$, $\sin\theta = \dfrac{1}{\sqrt{3}}$
$\theta \approx 35.3°$	$\theta \approx 35.3°$

> BOTH GIVE THE SAME NUMBER, AND BOTH ARE WORTH FULL MARKS
>
> The synthetic method needs a good drawing; the vector method needs good coordinates. Choose by which you can set up faster — but check the vector answer against a rough sketch, because a sign error in n is invisible in the algebra and obvious in the picture.

Distances

DISTANCE	HOW
point to plane	the perpendicular; or $\dfrac{\lvert n\cdot p - d\rvert}{\lvert n\rvert}$
point to line	the perpendicular in the plane containing both
parallel planes	any point of one to the other
skew lines	the common perpendicular



three orthogonal projections determine the solid

Every distance in space is the length of one perpendicular.

Standard results in a cube of edge a.

DISTANCE	VALUE
between two opposite faces	a
between two skew edges	a
from a vertex to the far space diagonal	$\frac{a\sqrt{6}}{3}$
from the centre to a face	$\frac{a}{2}$
from a vertex to the plane A_1BD	$\frac{a\sqrt{3}}{3}$

THE VOLUME TRICK FOR A POINT-TO-PLANE DISTANCE

For a tetrahedron, $V = \frac{1}{3}Bh$ gives $h = \frac{3V}{B}$. Computing the volume two ways — once with an easy base, once with the awkward one — finds the distance without any construction at all. It is the fastest route in almost every cube question.

Choosing a method

THE QUESTION GIVES	PREFER	BECAUSE
a cube or a box	vectors	the coordinates write themselves
a proof, no numbers	synthetic	the criteria are the argument
a regular pyramid	synthetic	one right triangle answers it
a plane by an equation	vectors	the normal is already there
skew lines	vectors	the common perpendicular is hard to draw
a distance in a cube	the volume trick	no construction needed

ONE SENTENCE TO REMEMBER THE WHOLE REVISION BY

Every three-dimensional problem is solved by finding the one right triangle it contains — or, in vector form, the one dot product it depends on. Two methods, one idea: reduce the space question to something two-dimensional.

02 Worked examples

EXAMPLE 1 **In a cube of edge 1, find the angle between AC_1 and the base, both ways.**

1. Synthetic: projection $AC = \sqrt{2}$, height 1.

   $$\tan\theta = \frac{1}{\sqrt{2}}$$

2. Vector: $d = (1,1,1)$, $n = (0,0,1)$.

3. $$\sin\theta = \frac{1}{\sqrt{3}}$$

4. $\theta \approx 35.3°$

   Both agree.

ANSWER

$\approx 35.3°$

EXAMPLE 2 **In a cube of edge a, find the distance from A to the plane A_1BD.**

1. Tetrahedron AA_1BD: $V = \frac{1}{6}a^3$.

   Three mutually perpendicular edges.

2. Face A_1BD is equilateral of side $a\sqrt{2}$.

3. $B = \frac{\sqrt{3}}{4}(2a^2) = \frac{a^2\sqrt{3}}{2}$

4. $h = \frac{3V}{B} = \frac{a}{\sqrt{3}} = \frac{a\sqrt{3}}{3}$

   The volume trick.

ANSWER

$\frac{a\sqrt{3}}{3} \approx 0.577a$

EXAMPLE 3 **Find the angle between the planes $x + y = 1$ and $y + z = 1$.**

1. $n_1 = (1,1,0)$, $n_2 = (0,1,1)$

2. $n_1 \cdot n_2 = 1$

3. $\cos\theta = \frac{1}{2}$

4. $\theta = 60°$

ANSWER

$60°$

EXAMPLE 4 **A point is *12* cm from a plane. Two obliques of *13* and *20* cm are drawn. Compare their projections and the angles they make with the plane.**

1. Projections 5 and 16.
   $\sqrt{169 - 144}$ and $\sqrt{400 - 144}$.

2. $sin\ \theta_1 = \dfrac{12}{13} \Rightarrow \theta_1 \approx 67.4°$

3. $sin\ \theta_2 = \dfrac{12}{20} \Rightarrow \theta_2 \approx 36.9°$

4. The longer oblique makes the smaller angle.

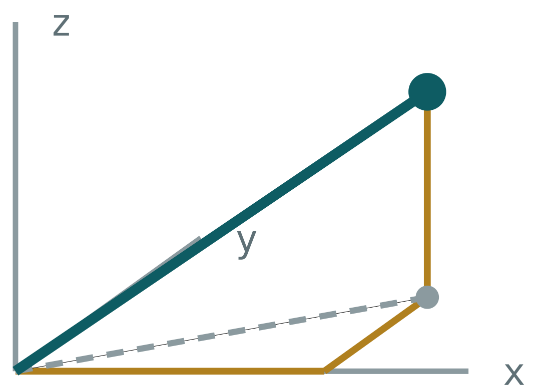
ANSWER

5 and 16 cm; $\approx 67.4°$ and $\approx 36.9°$

03 Interactive model

Set up the cube with coordinates on the board and solve one question twice; the class times both methods.

The cube, its diagonals and its planes



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Skew lines	Ayqash toʻgʻri chiziqlar	Скрещивающиеся прямые
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Perpendicular to a plane	Tekislikka perpendikulyar	Перпендикулярная плоскости
Three perpendiculars theorem	Uch perpendikulyar teoremasi	Теорема о трёх перпендикулярах
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Linear angle	Chiziqli burchak	Линейный угол
Orthogonal projection	Ortogonal proyeksiya	Ортогональная проекция
Synthetic method	Sintetik usul	Синтетический метод
Vector method	Vektor usuli	Векторный метод
Common perpendicular	Umumiy perpendikulyar	Общий перпендикуляр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A line ⊥ to a plane is ⊥ to:

- one line of it
- two intersecting lines of it, and then all
- two parallel lines
- the normal

The angle to a plane uses the vector formula with:

- *cos*
- *sin*
- *tan*
- *cot*

The distance from a vertex of a cube of edge a to the plane A_1BD:

- $\frac{a}{2}$
- $\frac{a\sqrt{3}}{3}$
- $\frac{a\sqrt{2}}{2}$
- a

For a cube question, the faster method is usually:

- synthetic
- vectors
- trial
- neither

For a proof with no numbers, prefer:

- vectors
- the synthetic criteria
- coordinates
- estimation

The longer oblique from a point makes:

- the larger angle
- the smaller angle
- the same angle
- a right angle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Three positions of two lines in space

   ANSWER Intersecting, parallel, skew

2. A line ⊥ to two intersecting lines of a plane is:

   ANSWER Perpendicular to the plane

3. Space diagonal of a cube of edge a

   ANSWER $a\sqrt{3}$

4. Face diagonal of the same

   ANSWER $a\sqrt{2}$

5. Projection of a 13 oblique from a point 12 away

   ANSWER 5

6. Distance between two opposite faces of a cube of edge a

   ANSWER a

7. Normal of $x + y = 1$

   ANSWER $(1, 1, 0)$

Medium · the standard question

7 PROBLEMS

1. Angle between AC_1 and the base of a cube

ANSWER $\approx 35.3°$

2. Angle between the planes $x + y = 1$ and $y + z = 1$

ANSWER $60°$

3. Distance from a vertex of a cube of edge a to the plane A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

4. Angles of obliques 13 and 20 from a point 12 from a plane

ANSWER $\approx 67.4°$, $\approx 36.9°$

5. Angle between two space diagonals of a cube

ANSWER $\approx 70.5°$

6. Distance between two skew edges of a cube of edge a

ANSWER a

7. Dihedral angle ABC_1 with the base of a cube

ANSWER $45°$

Hard · stretch and reason

7 PROBLEMS

1. Distance from a vertex of a cube of edge a to the far space diagonal

   ANSWER $\frac{a\sqrt{6}}{3}$

2. A regular tetrahedron of edge a: the angle between a face and the base

   ANSWER $\approx 70.5°$

3. Same tetrahedron: the distance from a vertex to the opposite face

   ANSWER $\frac{a\sqrt{6}}{3}$

4. A regular square pyramid, base 12, height 8: the dihedral angle along a base edge

   ANSWER $\approx 53.1°$

5. Same pyramid: the angle of a lateral edge with the base

   ANSWER $\approx 43.3°$

6. Shortest distance between $r = t(1,0,0)$ and $r = (0,1,1) + s(0,1,0)$

   ANSWER 1

7. Prove that the three face diagonals from one vertex of a cube are not coplanar

   ANSWER Their scalar triple product is $-2a^3 \neq 0$

07 Homework

Homework — 5 tasks

Solve at least two questions by both methods and record which was faster.

1. State the five criteria of the first table, each with a sketch.
2. In a cube of edge 6 cm, find the angle between the space diagonal and the base by both methods.
3. Find the angle between the planes $2x - y + z = 3$ and $x + y + 2z = 1$.

4. A point is 9 cm from a plane; obliques of 15 and 41 cm are drawn. Find both projections and both angles.
5. Find the distance from a vertex of a cube of edge 4 cm to the plane through the three adjacent vertices, using the volume trick.